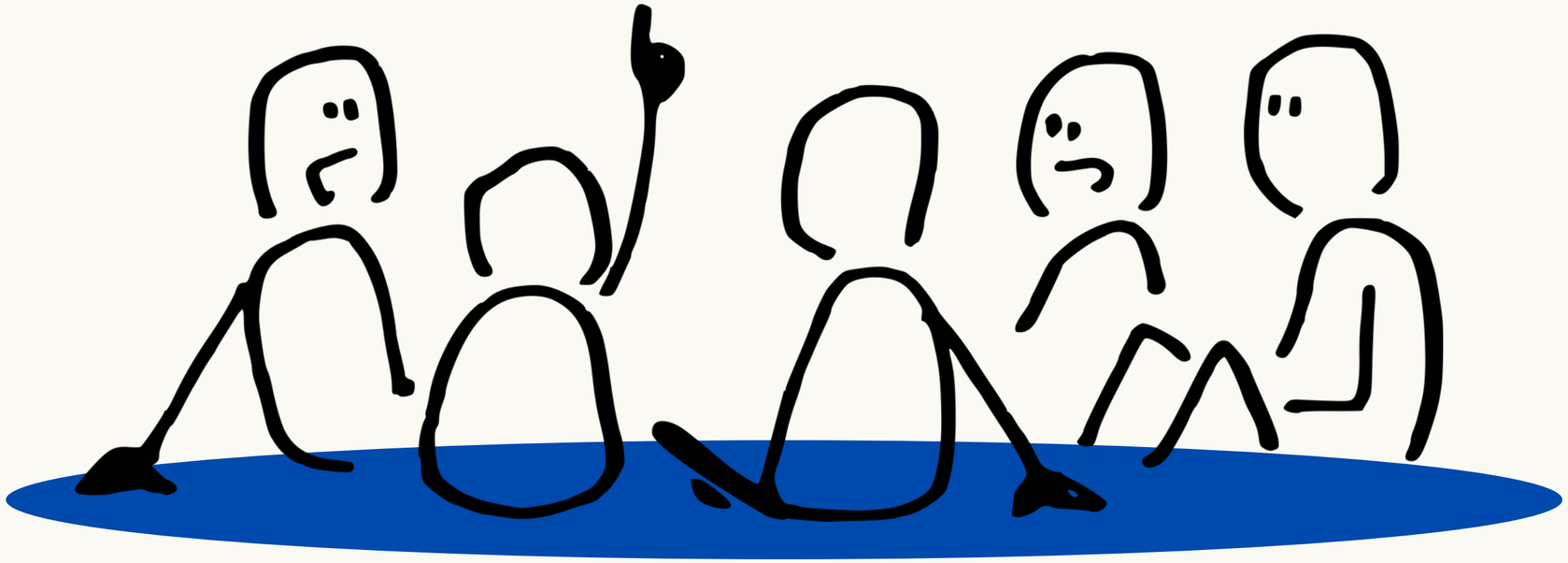
From Reviews to Insights:

SENTIMENT ANALYSIS AND EXTRACTIVE SUMMARIZATION OF BALI HOTEL REVIEWS



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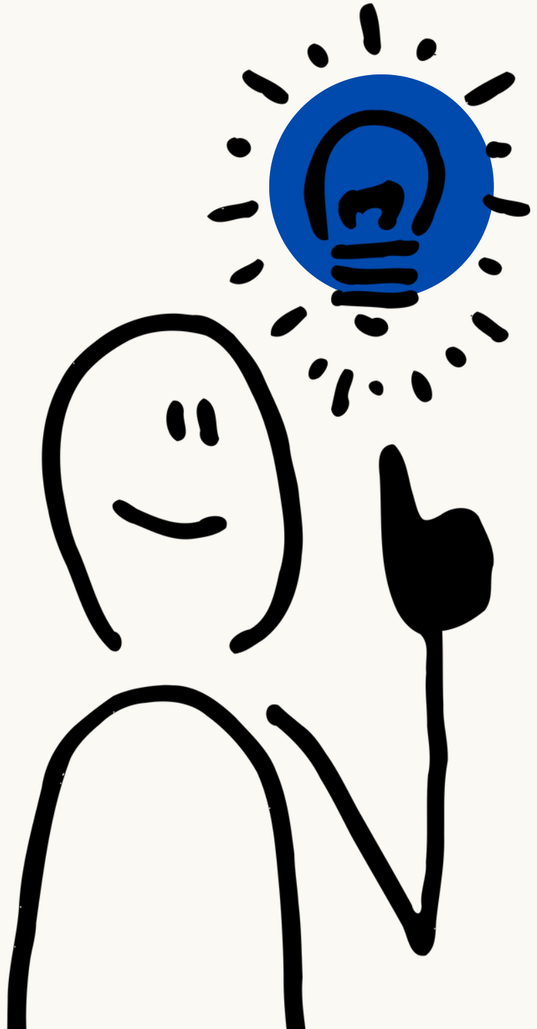
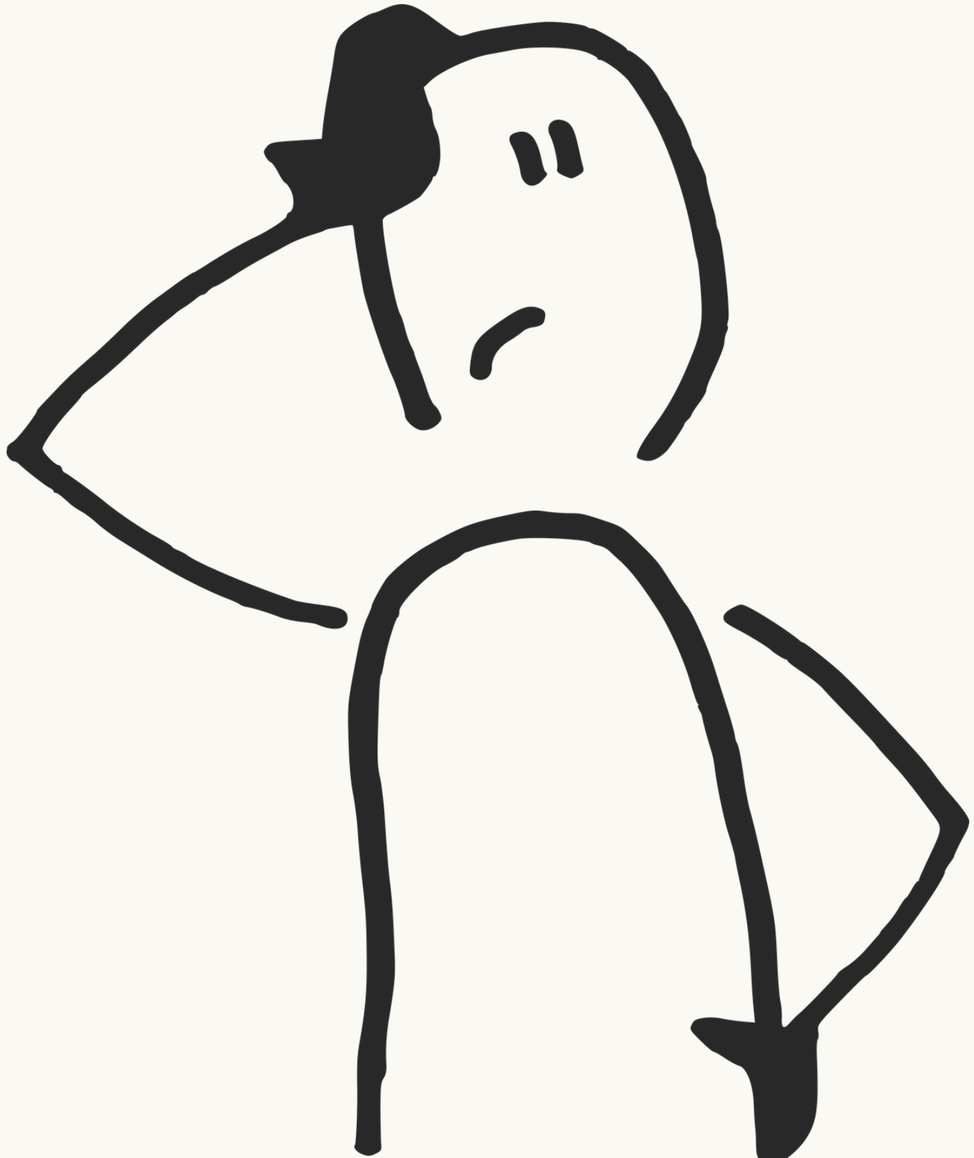
Tourism is an important sector in Indonesia, with Bali being one of the most popular destinations due to its natural beauty, culture, and hospitality.



Bali attracts the world, but hotels convince them to stay.

Online reviews now act as the global megaphone for these experiences, guiding travelers' expectations and decisions across the world.

PROBLEM



Visitor Satisfaction

The quality of hotel and hospitality services is crucial in shaping visitor experiences in Bali, a top tourism destination in Indonesia.

Influence of Reviews

Millions of travelers share opinions online, and these reviews strongly impact potential visitors' booking decisions and perceptions of hotel quality.

Analysis Challenge

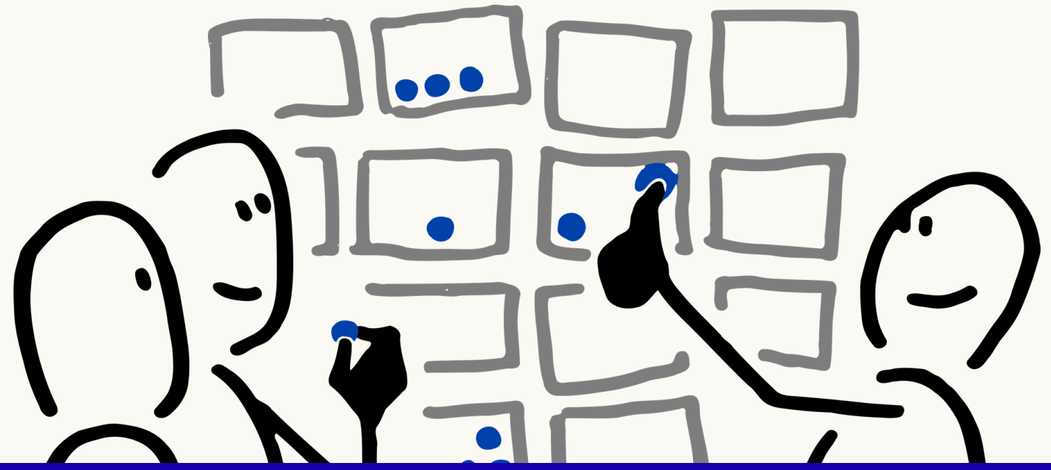
The vast volume of online reviews makes it difficult for hotel managers to manually analyze feedback, identify trends, and extract actionable insights to improve service quality and maintain competitiveness.

PROPOSED APPROACH

“Online reviews hold valuable insights, and sentiment analysis unlocks them”

By transforming thousands of guest comments into clear, actionable insights, sentiment analysis empowers hotels in Bali to understand what truly shapes visitor satisfaction, quickly spot service gaps, enhance operations, and deliver experiences that keep tourists returning.

DATASET PREPARATION



Dataset Overview

The dataset used in this project consists of 3,017 hotel reviews scraped from an online travel platform, containing 24 fields such as hotel information, reviewer details, review text, rating, language, and location metadata.

Text Preprocessing

The Review column contains raw, unstructured text that undergoes preprocessing steps, such as tokenization, noise removal, and lowercasing, to clean and standardize the text before it is used for sentiment analysis

Label Creation

The Rating column is converted into a binary Label with two sentiment classes: Positive (rating > 3) and Negative (rating ≤ 3). This conversion simplifies the task into a two-class sentiment classification problem.

For sentiment analysis, the key fields utilized are the hotel name, review text, and rating.

DeBERTaV3 MODEL

Why DeBERTaV3?

DeBERTaV3 is selected for this task as it is one of the top-performing models for sentiment analysis, offering strong contextual understanding, improved attention mechanisms, and state-of-the-art accuracy on text classification benchmarks.

Review Summarization

- Reviews are first grouped based on each hotel.
- A pre-trained BART summarization model (from Hugging Face) is used to generate concise summaries for every hotel.
- These summaries capture the main themes and recurring points found in the reviews.
- The summarized outputs will be used as inputs for the Aspect Extraction and Sentiment Classification step.

ASPECT EXTRACTION & SENTIMENT CLASSIFICATION

Aspect Extraction

Summaries are processed with spaCy using POS tagging.

Feature Identification

Nouns representing hotel features (e.g., bed, food, service) are extracted as aspects.

Sentiment Classification

Each aspect's sentence is classified using the trained model.

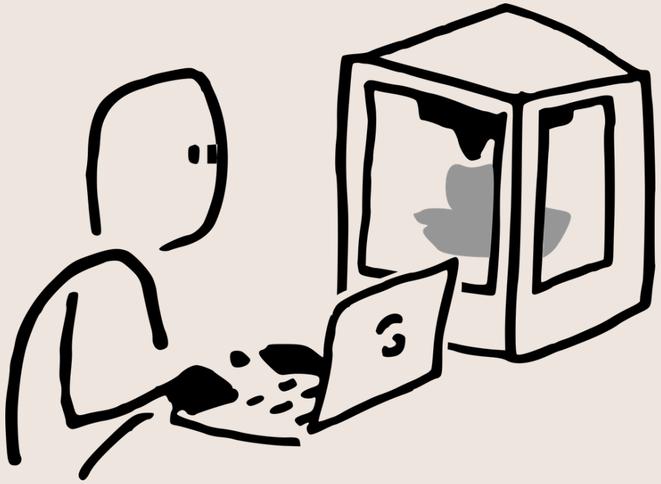
Polarity Output

Final results present sentiment for each aspect, e.g., bed: positive, food: negative.

INTEGRATION & OUTPUT

Streamlit Interface

Users enter a hotel name through a simple UI.

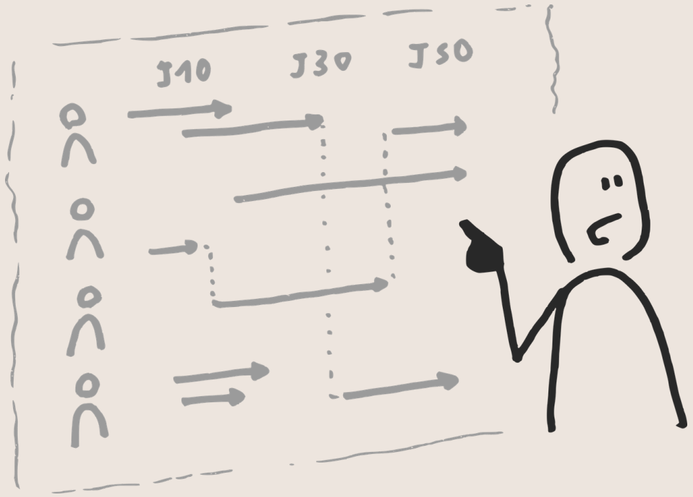


Automated Processing

The system generates a hotel-level summary and aspect sentiments.

Result Display

Summaries and sentiment outputs are presented clearly to the user.

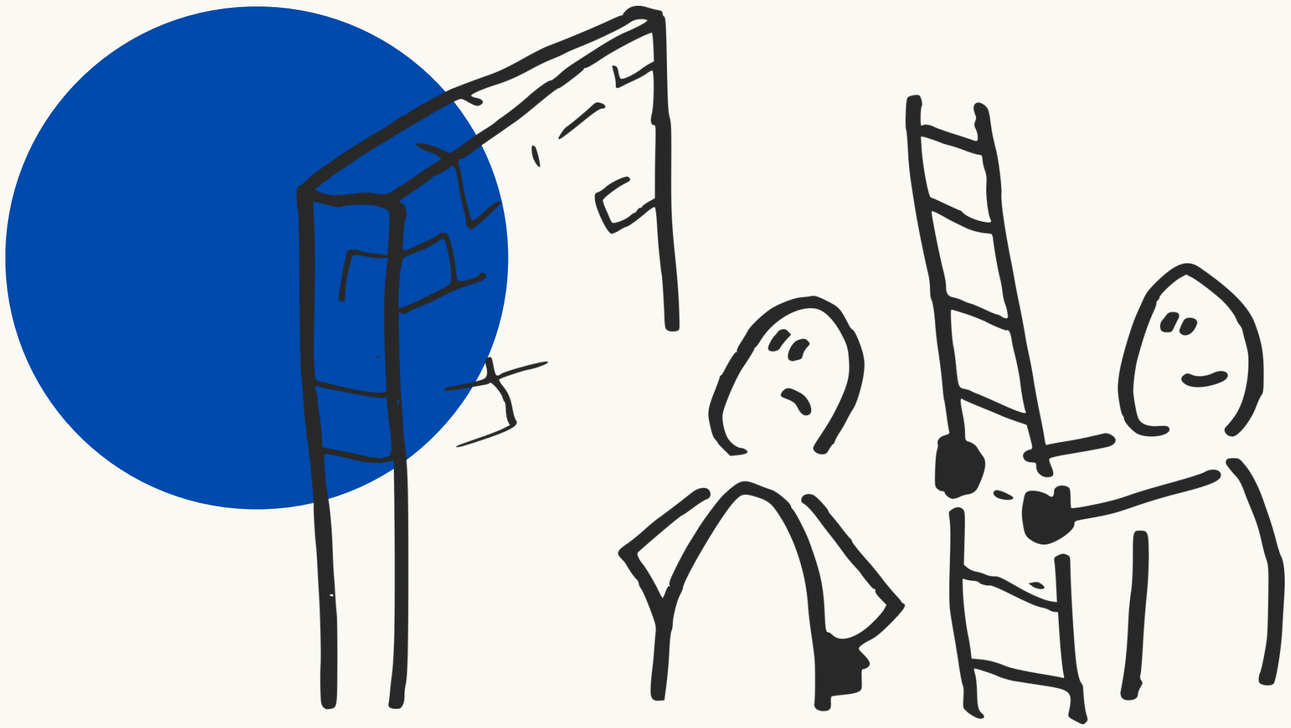


EVALUATION

The evaluation process is designed to assess both the quality of the generated summaries and the performance of the sentiment classification model.

Model Assessment

Sentiment model performance is measured using accuracy, precision, recall, and F1-score.



Summary Evaluation

Summaries are assessed using ROUGE scores and BERTScore.

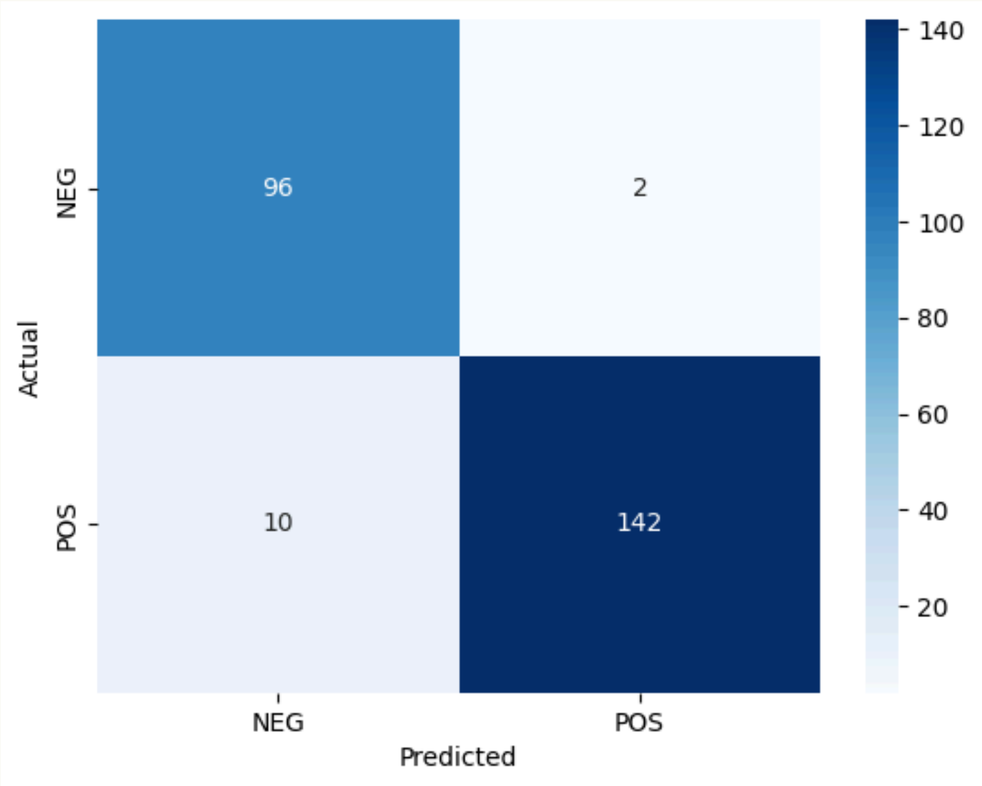
RESULTS

Accuracy	Precision	Recall	F1-Score
95.20%	98.61%	93.42%	95.94%

Overall, the model achieves strong results across all evaluation metrics.

- The DeBERTaV3 model achieves an **accuracy** of 95.20%, indicating that it correctly classifies the sentiment of the majority of customer reviews.
- **Precision** is the highest metric at 98.61%, indicating effective identification of positive sentiment.
- **Recall** is 93.42%, the lowest metric, but still demonstrates reliable sentiment detection.
- An **F1-score** of 95.94% confirms the model's robustness and balance.

Confusion Matrix



Out of 250 test samples, only 12 misclassifications occurred (2 false positives, 10 false negatives).

Misclassifications are mainly caused by complex contexts such as **sarcasm**, which can mislead the model's interpretation.

ROUGE & BERT SCORE

Accuracy	Precision
ROUGE-1	0.1099
ROUGE-2	0.0180
ROUGE-L	0.0712
BERT SCORE	0.8202

PYTHON CODE EXPLANATION

1 Library Imports & Setup

- Imports all required libraries for data handling, visualization, NLP processing, model training, summarization, aspect extraction, and evaluation.
- NLTK resources are downloaded to support tokenization and stopword removal.

2 Data Loading

- Loads the scraped hotel review dataset from a CSV file.
- Displays basic information to understand the dataset structure.

3 Data Cleaning & Selection

- Renames columns to improve readability (e.g., hotel name).
- Keeps only relevant columns: Hotel, rating, review text, and language.
- Removes missing values and duplicate records.
- Filters reviews to English-only for consistent NLP processing.

4 Exploratory Data Analysis (EDA)

- Analyzes hotel distribution to see how reviews are spread across hotels.
- Visualizes hotel frequency and rating distribution using bar charts.
- Helps understand class balance and dataset characteristics.

PYTHON CODE EXPLANATION

5 Sentiment Label Creation

- Converts numerical ratings into sentiment labels:
  - Positive: rating > 3
  - Negative: rating ≤ 3
- Encodes labels into numerical values (0 = Negative, 1 = Positive) for model training.

6 Text Preprocessing

- Cleans review text by converting to lowercase, removing URLs, HTML tags, special characters, and extra spaces
- Produces a standardized clean_text column suitable for transformer models.

7 Dataset Splitting

- Splits the data into: 80% training, 10% validation, and 10% testing
- Uses stratified sampling to preserve sentiment distribution across splits.

8 Hugging Face Dataset Conversion

- Converts Pandas DataFrames into Hugging Face Dataset format.
- Organizes datasets into a DatasetDict for training, validation, and testing.

9 Tokenization (DeBERTaV3)

- Loads the DeBERTaV3 tokenizer.
- Tokenizes cleaned reviews with padding, truncation, and max length of 256 tokens
- Removes unnecessary columns and renames labels for compatibility with Hugging Face Trainer.

PYTHON CODE EXPLANATION

10 Model Initialization

- Loads DeBERTaV3-base for binary sentiment classification.
- Configures the model with two output classes: Positive and Negative.

11 Training Configuration

- Defines training parameters:
  - Learning rate, batch size, epochs
  - Weight decay and early stopping
  - Uses EarlyStoppingCallback to prevent overfitting.
- Saves the best-performing model.

12 Model Training

- Trains the DeBERTaV3 model using the training dataset.
- Evaluates performance on the validation set after each epoch.

13 Model Evaluation

- Evaluates performance using Accuracy, Precision, Recall, and F1-score
- Generates predictions on the test set.
- Displays a confusion matrix and classification report.
- Analyzes misclassifications (e.g., sarcasm and contextual ambiguity)

PYTHON CODE EXPLANATION

14 Model Saving & Loading

- Saves the trained sentiment model and tokenizer.
- Compresses and downloads the model for reuse.
- Reloads the model for inference tasks.

15 Review Grouping by Hotel

- Groups reviews by hotel name.
- Combines all reviews per hotel into a single text block.
- Prepares data for hotel-level summarization.

16 Extractive Summarization

- Uses manual extractive summarization.
- Selects important sentences based on word frequency.
- Serves as a baseline for comparison with BART summaries.

17 Abstractive Summarization (BART)

- Applies BART (facebook/bart-large-cnn) for abstractive summarization.
- Generates concise hotel-level summaries.
- Handles long input safely with truncation.

PYTHON CODE EXPLANATION

18 Summarization Evaluation

- Uses ROUGE scores to compare extractive and BART summaries.
- Uses BERTScore to measure semantic similarity.
- Quantifies summary quality both lexically and semantically.

19 Aspect Extraction (spaCy)

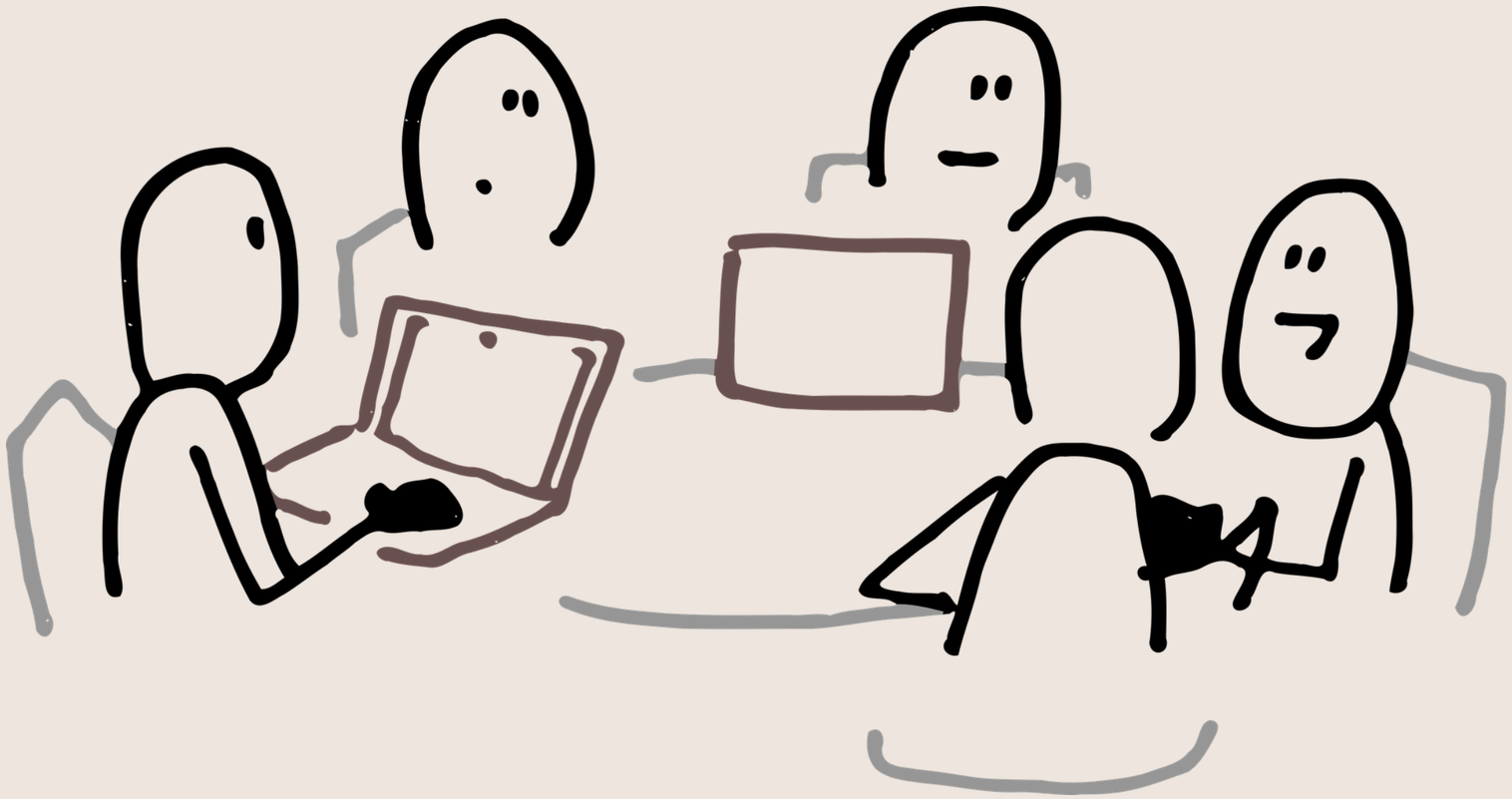
- Uses spaCy POS tagging to extract noun-based hotel aspects.
- Filters aspects using a predefined hotel-related vocabulary.
- Normalizes terms using lemmatization.

20 Aspect-Based Sentiment Classification

- Extracts sentences related to each aspect.
- Applies the trained DeBERTaV3 model to classify sentiment per aspect.
- Uses majority voting when multiple sentences mention the same aspect.
- Produces outputs like:
  - room: positive
  - food: negative

21 Streamlit Application

- Interactive web interface using Streamlit.
- Allows users to input hotel reviews, view summarized reviews, and see aspect-based sentiment results.

THANK YOU

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